

# TECHNICAL REPORT

## Bilingual Telugu + English Automatic Speech Recognition using Whisper

*Project Progress Report – Training, Evaluation, Error Analysis and Inference*

**Colab link:** <https://colab.research.google.com/drive/1KhBcZKNGVaPy91J-iT-pe-TEw-VWlu7I?usp=sharing>

**GITHUB Link:** <https://github.com/EmaniSaketh/Whisper-base-26hours.git>

### 1. Project Overview

This project focuses on developing a bilingual Automatic Speech Recognition (ASR) system for Telugu and English speech. The approach used is fine-tuning the OpenAI Whisper Base model on a combined Telugu and English speech dataset. The objective is to adapt the pre-trained Whisper model to the target bilingual speech data and evaluate its transcription performance using Word Error Rate (WER), Character Error Rate (CER), and qualitative error analysis.

### 2. Model Used

The model selected for the current implementation is Whisper Base. Whisper is available in multiple model sizes including Tiny, Base, Small, Medium, Large, Large-v2 and Large-v3. For this project, the Base model was selected and fine-tuned rather than using the larger variants.

The final trained model is represented by checkpoint-3915. This checkpoint is not a separate Whisper model size; it is a training checkpoint containing the fine-tuned version of the Whisper Base model.

### 3. Training Dataset

The training data consisted of Telugu and English speech. The target training composition was approximately 15 hours of Telugu and 15 hours of English speech. The available prepared datasets contained approximately 11.90 hours of Telugu training audio and 15.00 hours of English training audio, giving approximately 26.90 hours of combined training data.

| Dataset           | Audio                                  | Approx. Duration | Status            |
|-------------------|----------------------------------------|------------------|-------------------|
| Telugu Training   | 7,216 WAV files / 7,215 paired records | 11.90 hours      | Prepared          |
| English Training  | Prepared English training set          | 15.00 hours      | Prepared          |
| Combined Training | Telugu + English                       | 26.90 hours      | Used for training |
| Telugu Validation | 903 WAV + 903 TXT                      | 1.5 hours        | Evaluated         |
| English Test      | 713 FLAC + 713 TXT                     | 1.5 hours        | Evaluated         |

### 4. Training Configuration and Progress

The Whisper Base model was fine-tuned using the bilingual Telugu and English training data. Training was completed for 3 epochs and 3,915 training steps. The training process was run using an NVIDIA T4 GPU.

The final reported validation loss at the end of training was 0.104132. Training checkpoints were saved during the process so that the trained model could be recovered and evaluated.

| Training Item         | Value                                       |
|-----------------------|---------------------------------------------|
| Base Model            | Whisper Base                                |
| Languages             | Telugu + English                            |
| Training Duration     | Approximately 26.90 hours of prepared audio |
| Epochs                | 3                                           |
| Training Steps        | 3,915                                       |
| Training GPU          | NVIDIA T4                                   |
| Final Validation Loss | 0.104132                                    |
| Final Checkpoint      | checkpoint-3915                             |

## 5. Saved Checkpoints

The following training checkpoints were saved to Google Drive:

- checkpoint-3000
- checkpoint-3500
- checkpoint-3915

The final checkpoint, checkpoint-3915, contains the fine-tuned model files, tokenizer and training state information. The checkpoint was successfully loaded for inference and evaluation.

## 6. English Evaluation

The English test set contained 713 audio samples with corresponding reference transcripts. The complete English evaluation was performed using the final checkpoint.

| Metric                     | Result |
|----------------------------|--------|
| Test Samples               | 713    |
| Word Error Rate (WER)      | 6.07%  |
| Character Error Rate (CER) | 2.03%  |

The complete English evaluation results were saved as english\_evaluation\_results.csv in Google Drive. The file contains the audio filename, reference transcription and model prediction for all 713 samples.

## 7. Telugu Evaluation

The Telugu validation set contained 903 audio samples, with 903 corresponding text files. The final checkpoint was evaluated on all 903 samples.

| Metric                | Result                   |
|-----------------------|--------------------------|
| Validation Samples    | 903                      |
| Word Error Rate (WER) | 45.25%                   |
| Evaluation Runtime    | Approximately 33 minutes |

The Telugu evaluation used Whisper decoding configured for the Telugu language and transcription task. The evaluation completed successfully for all 903 samples.

## 8. Real-Audio Inference Examples

Real audio inference was also performed using the final fine-tuned checkpoint. The following examples demonstrate how the model behaves on individual Telugu and English recordings.

## 8.1 Telugu Example

Audio file: 000010042.wav

Reference: చింటూ అలియాస్ చంద్రశేఖర్ చంద్రశేఖర్ నివాసంలో పోలీసులు సోదా చేసి

Prediction: చింటూ అల్యాస్ చంద్రశేఖర్ నివాసను పోలీసులు సోదా చేసి

Observed differences included a spelling variation in the word corresponding to 'alias', one omitted occurrence of 'చంద్రశేఖర్', and a difference between 'నివాసంలో' and 'నివాసను'. The remaining words in the example were transcribed correctly.

## 8.2 English Example

Audio file: 1580-141083-0024.flac

Reference: YOU LEFT HIM IN A CHAIR YOU SAY WHICH CHAIR BY THE WINDOW THERE

Prediction: YOU LEFT HIM IN A CHAIR YOU SAY WHICH CHAIR BY THE WINDOW THERE

The prediction exactly matched the reference transcription for this example.

## 9. Error Analysis

Error analysis was performed to understand the differences between reference transcripts and model predictions. WER was used as the primary ASR evaluation metric, while CER was also calculated for the English evaluation.

The English results show a WER of 6.07% and CER of 2.03%. The Telugu validation results show a WER of 45.25%. The individual Telugu example demonstrates errors involving word omission and small lexical/spelling differences.

A detailed substitution, deletion and insertion breakdown can be calculated from the saved evaluation predictions. It has not been included as a numerical result in this report because the final verified S/D/I counts were not part of the completed evaluation record used here.

## 10. Current Project Status

- Whisper Base selected as the base ASR model.
- Bilingual Telugu + English training dataset prepared.
- Approximately 26.90 hours of prepared training audio used.
- Fine-tuning completed for 3 epochs and 3,915 steps.
- Final checkpoint checkpoint-3915 saved and successfully loaded.
- English evaluation completed on 713 samples: WER 6.07%, CER 2.03%.
- Telugu evaluation completed on 903 samples: WER 45.25%.
- Real-audio inference demonstrated for both English and Telugu.
- English evaluation predictions saved to english\_evaluation\_results.csv.
- Training checkpoints and evaluation outputs are stored in Google Drive.

## 11. Mentor Demonstration / Evidence Available

For demonstration, the project can be shown directly in Google Colab. The saved checkpoint directory can be displayed, the final checkpoint can be loaded, and the saved English evaluation CSV can be opened to show the 713 predictions. Individual Telugu and English audio files can also be passed through the model to demonstrate real inference.

The checkpoint's `trainer_state.json` can be used to display the actual recorded training log, including training steps, epoch information, loss values, learning rate and evaluation loss, without fabricating or manually recreating training logs.

## 12. Conclusion

The current project successfully produced a fine-tuned bilingual Whisper Base ASR model for Telugu and English speech. Training was completed and the final checkpoint was preserved for reproducibility and inference. The model achieved 6.07% WER on the 713-sample English test set and 45.25% WER on the 903-sample Telugu validation set. Real-audio inference confirms that the fine-tuned model can generate meaningful transcriptions in both languages, while the evaluation results identify a clear difference in performance between the English and Telugu test sets.

## 13. Key Files and Paths

- `/content/drive/MyDrive/whisper_bilingual_checkpoints/checkpoint-3000`
- `/content/drive/MyDrive/whisper_bilingual_checkpoints/checkpoint-3500`
- `/content/drive/MyDrive/whisper_bilingual_checkpoints/checkpoint-3915`
- `/content/drive/MyDrive/english_evaluation_results.csv`
- `/content/drive/MyDrive/telugu_validation_1.5h/audio`
- `/content/drive/MyDrive/telugu_validation_1.5h/text`
- `/content/drive/MyDrive/english_test_1.5h/audio`
- `/content/drive/MyDrive/english_test_1.5h/text`